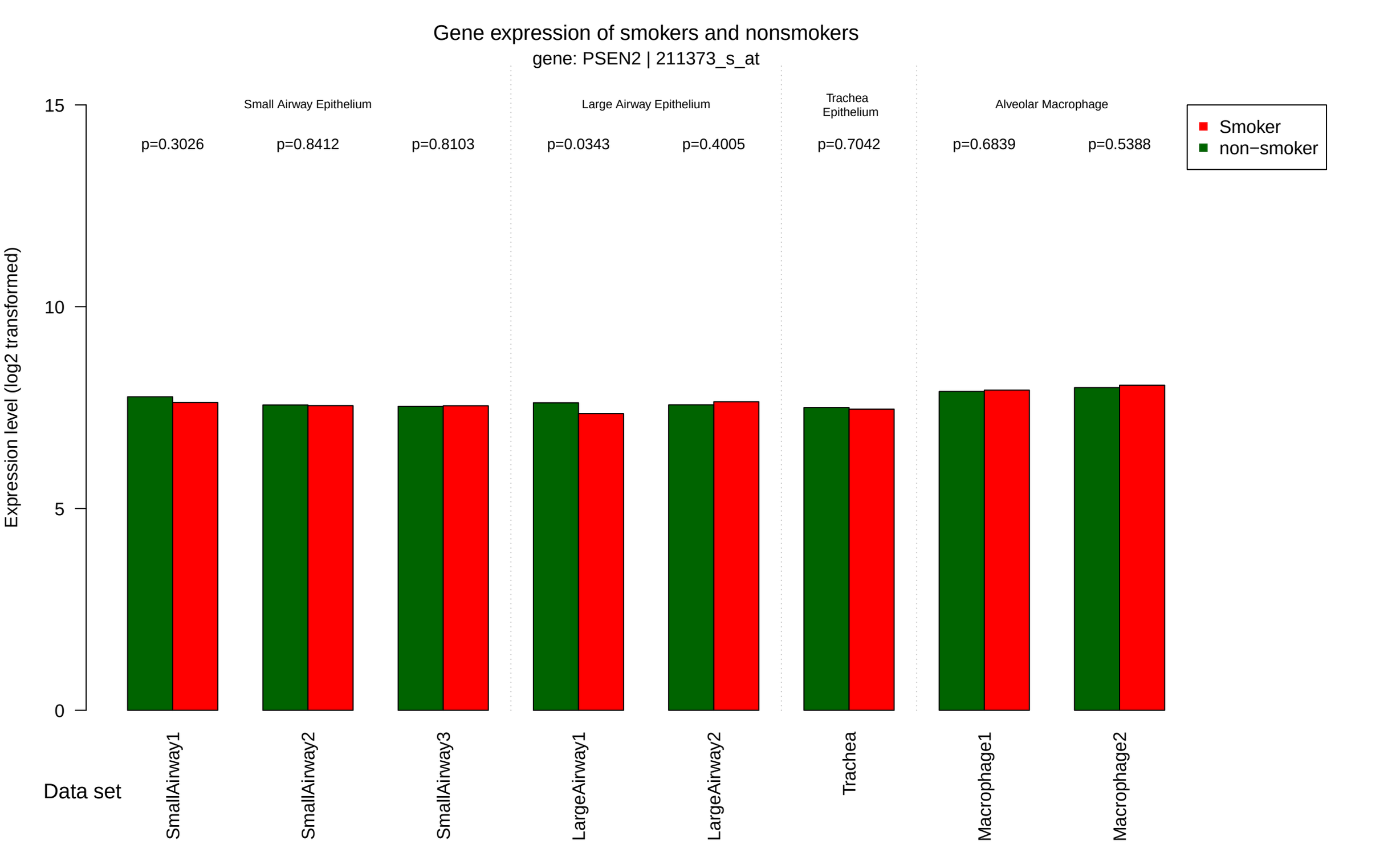
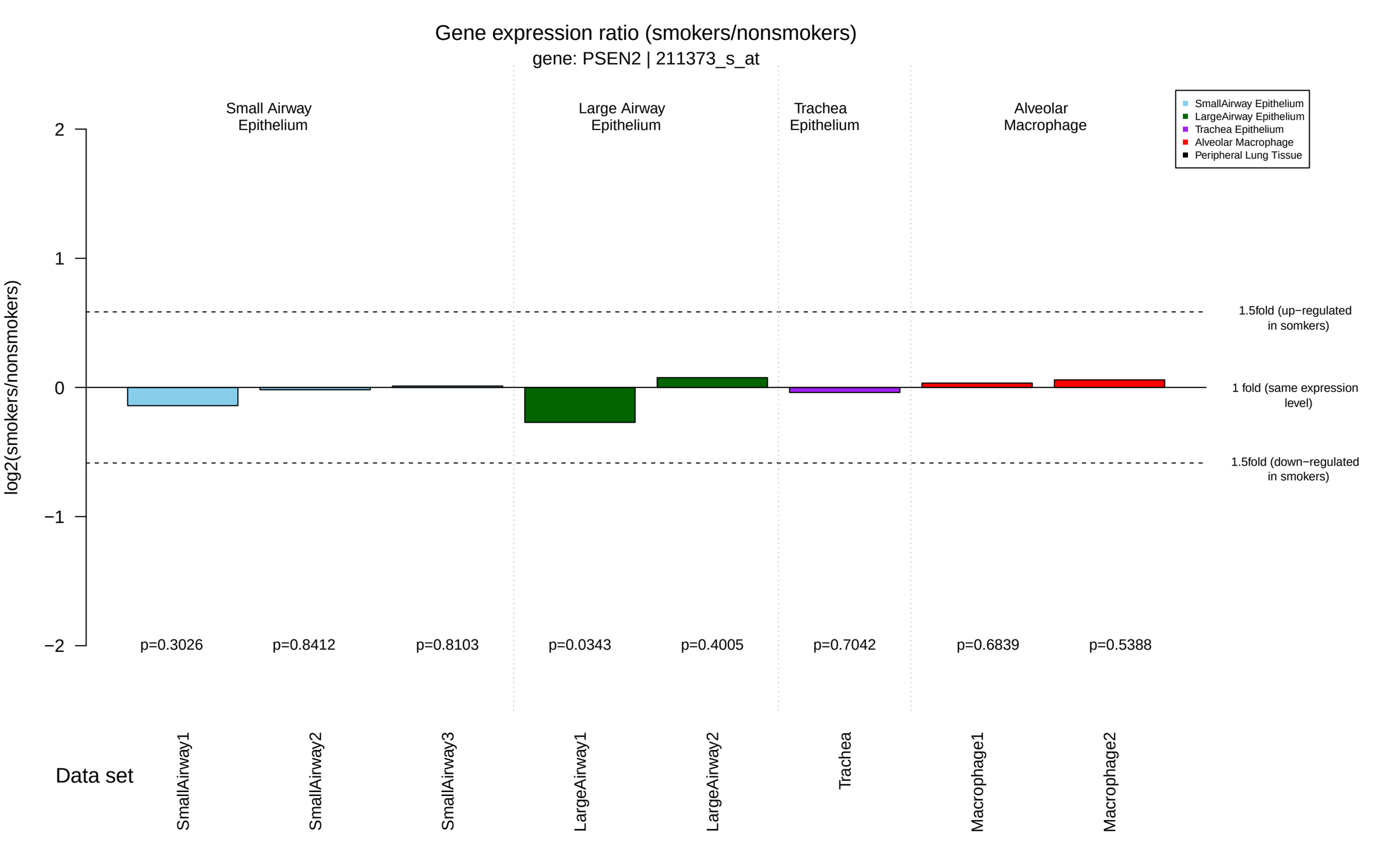
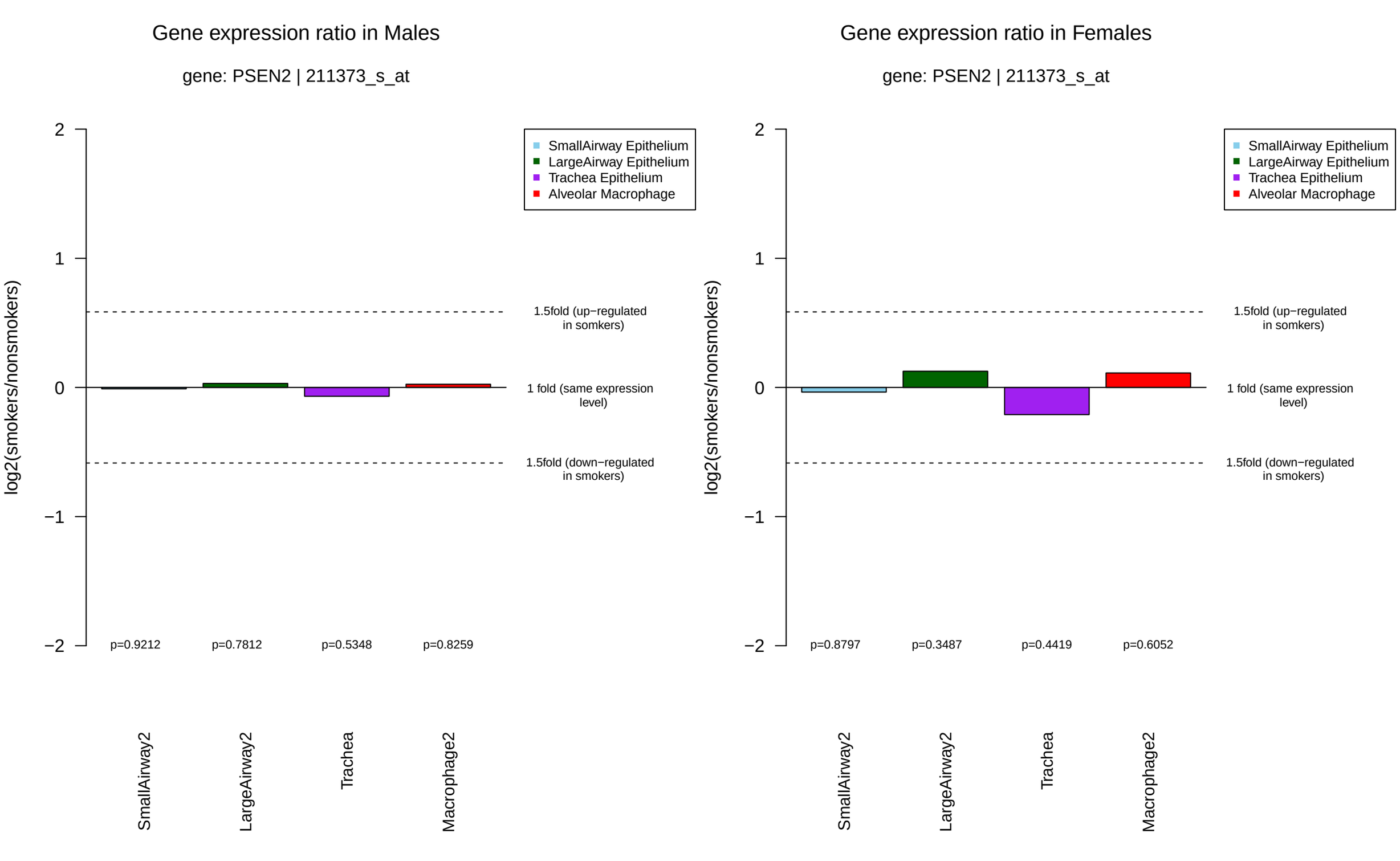
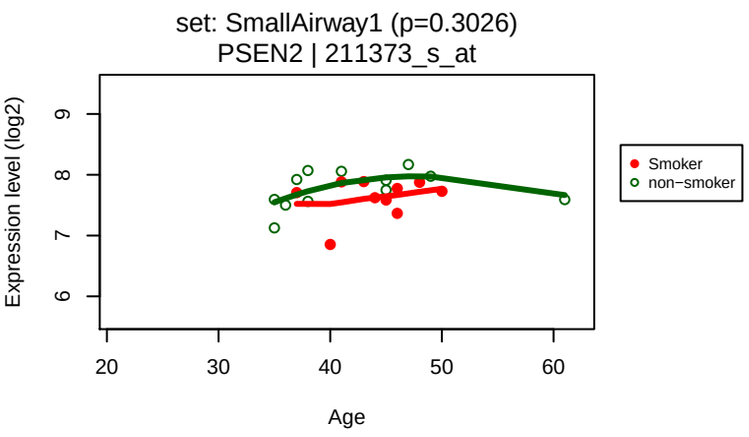
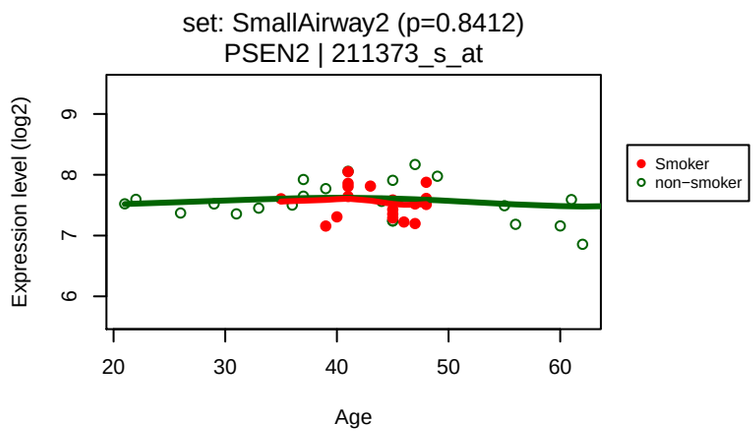
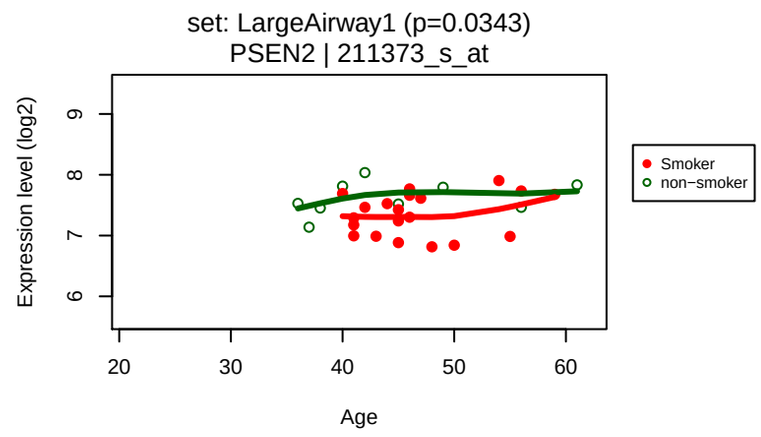

Gene expression of smokers and nonsmokers
gene: PSEN2 | 211373_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3026
p=0.8412
p=0.8103
p=0.0343
p=0.4005
p=0.7042
p=0.6839
p=0.5388
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

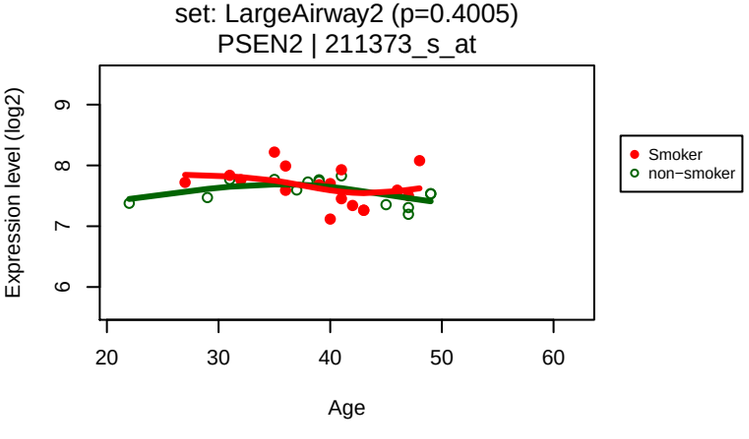

Gene expression ratio (smokers/nonsmokers)
gene: PSEN2 | 211373_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3026
p=0.8412
p=0.8103
p=0.0343
p=0.4005
p=0.7042
p=0.6839
p=0.5388
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

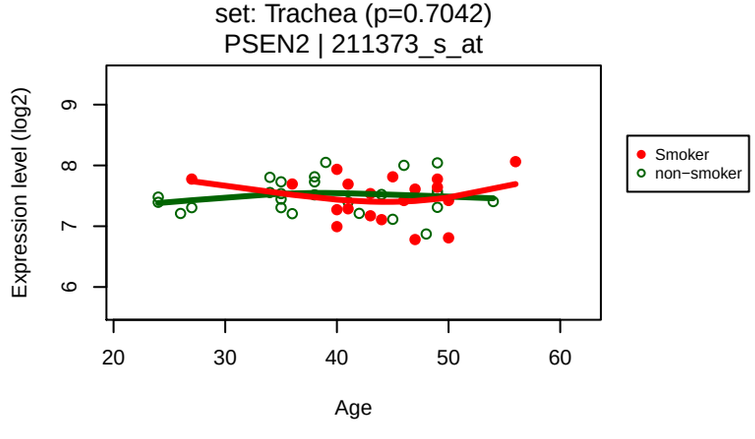

Gene expression ratio in Males
gene: PSEN2 | 211373_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9212
p=0.7812
p=0.5348
p=0.8259
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSEN2 | 211373_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8797
p=0.3487
p=0.4419
p=0.6052
SmallAirway2
LargeAirway2
Trachea
Macrophage2

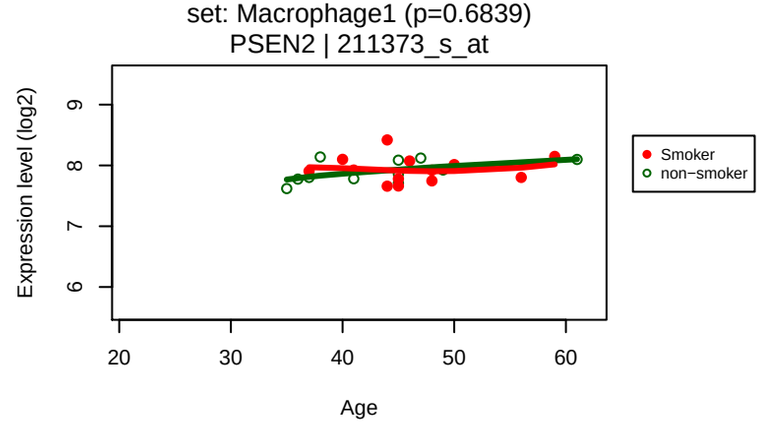

set: SmallAirway1 (p=0.3026)
PSEN2 | 211373_s_at
Expression level (log2)
Age
Smoker
non−smoker

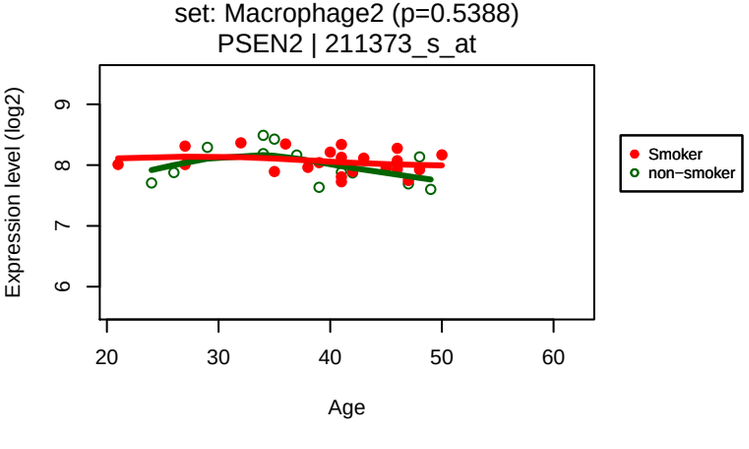

set: SmallAirway2 (p=0.8412)
PSEN2 | 211373_s_at
Expression level (log2)
Age
Smoker
non−smoker

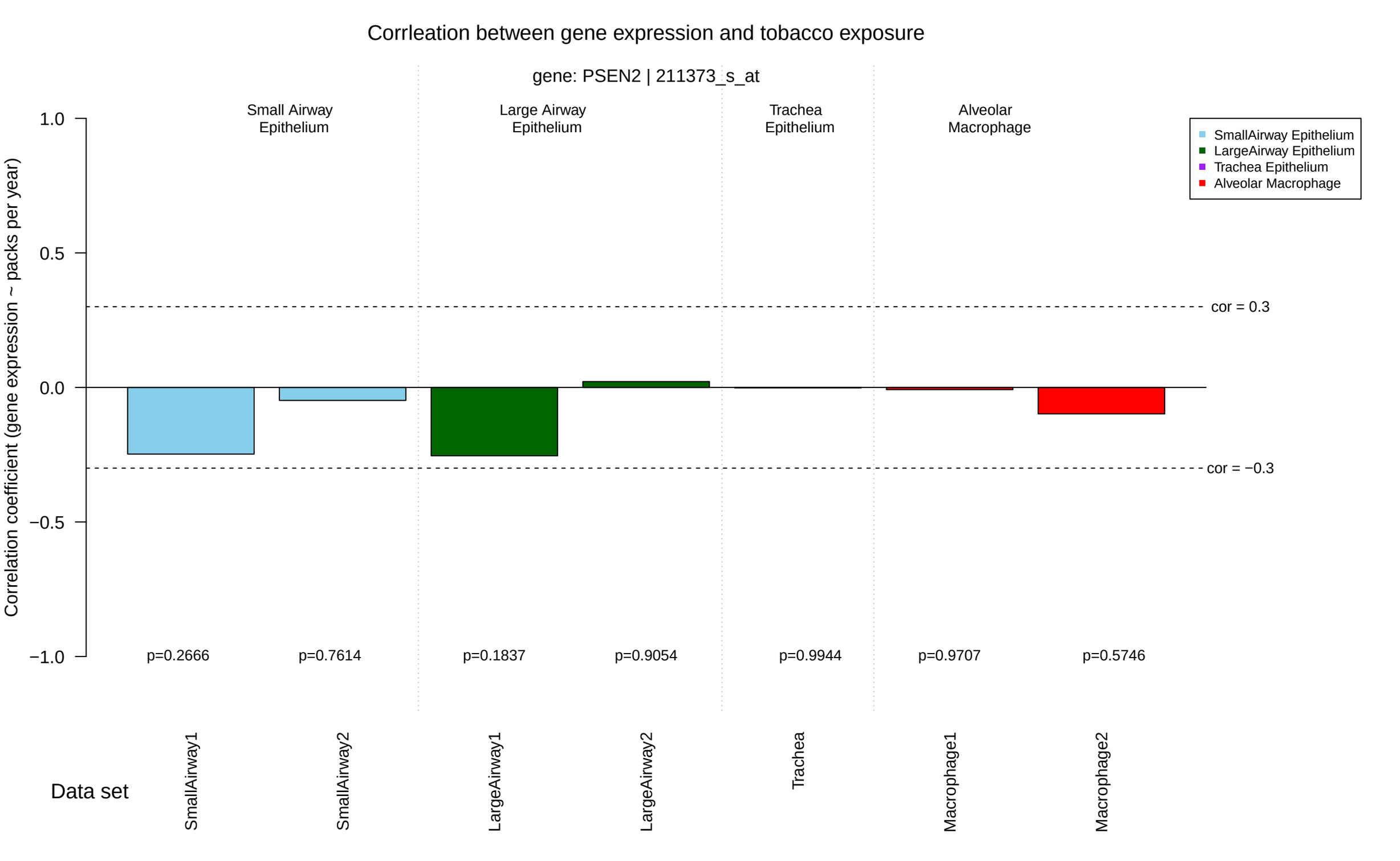

set: LargeAirway1 (p=0.0343)
PSEN2 | 211373_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4005)
PSEN2 | 211373_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7042)
PSEN2 | 211373_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6839)
PSEN2 | 211373_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5388)
PSEN2 | 211373_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSEN2 | 211373_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2666
p=0.7614
p=0.1837
p=0.9054
p=0.9944
p=0.9707
p=0.5746
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2